Test - III

1) What is rendering in React ? How does JSX renders to the screen?

Rending is the process of displaying UI in the browser.

Flow:

i) You write UI using JSX
(ii) JSX is converted into React.createElement()
(iii) React creates a virtual DOM
(iv) React compares change (diffing)
(v) Update only required parts in the Real DOM

2) Rendering a JavaScript variable inside JSX.

```
const name = "Kishore";
function App () {
    return <h1> Hello {name} </h1>;
}
```

3) Rendering object properties

```
const user = {
    name : "Kishore",
    age : 21
};
```

```
function App(){
    return(
      <div>
        <h1>{user.name}</h1>
        <p>{user.age}</p>
      </div>
    );
}
```

4) Rendering an array of values.

```
const fruits = ["Apple", "Banana", "Mango"];

function App(){
    return (
      <ul>
        {fruits.map((fruit, index) => (
        <li key={index}>{fruit}</li>
        ))}
      </ul>
    );
}
```

5) Rendering an array of objects using .map()

```
const users = [
    {id: 1, name: "kishore"},
    {id: 2, name: "Arun"}
];
```

```
function App() {
    return (
        <ul>
            {users.map(user => (
                <li key={user.id}> {user.name} </li>
            ))}
        </ul>
    );
}
```

6) Why is the key prop important?

* Identifies elements uniquely.
* Helps React track changes efficiently
* Prevents unnecessary re-renders
* Avoids UI bugs when list updates.

7) Rendering nested arrays (e.g., skills inside users)

```
const users = [
    { id: 1,
      name: "Kishore",
      skills: ["JS", "React"]
    }
],
    function App () {
```

```
return(
  <div>
    {users.map(user => (
      <div key={user.id}>
        <h2>{user.name}</h2>
        <ul>
          {user.skills.map((skill, index) => (
            <li key={index}>{skill}</li>
          ))}
        </ul>
      </div>
    ))}
  </div>
);
}
```

8) What is conditional rendering?

```
const isLoggedIn = true;
function App(){
  return (
    <div>
      {isLoggedIn? <h1>Welcome</h1>:
                   <h1>Please login</h1>}
    </div>
  );
}
```

9) Ternary operator in JSX

condition ? value1 : value2

Ex: {isloggedIn ? <Dashboard /> : <login />}

10) Difference && vs Ternary Operator && (logical AND)

{isloggedIn && <Dashboard />}

* Renders only if condition is true
* No else part

Ternary operator

{isloggedIn ? <Dashboard /> : <login />}